Chapter 10. Estimating Population Size and Nonresponse

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1 Capture–Recapture Estimation of N

To estimate the number N of fish in a lake, catch n_1 fish, mark them and release them; later catch a second sample of n_2 fish and count the number m of marked fish among them.

Definition 10.1: Assumptions

1. The population is closed (no births, deaths or migration between the samples).
2. Each sample is an SRS from the population (marking does not change catchability).
3. The two samples are independent.

The proportion of marked fish in the second sample should resemble that in the population:

$$\frac{m}{n_2} \approx \frac{n_1}{N} \quad \Longrightarrow \quad \hat{N} = \frac{n_1 n_2}{m}$$

(the Lincoln–Petersen estimator), where n_1 is the size of the first (capture) sample, n_2 that of the second (recapture) sample, and m the number of marked elements in the second sample.

1.1 Capture–recapture as a ratio estimator

Theorem 10.2

$\hat{N}$ is a ratio estimator of a population total, and

$$\widehat{\mathrm{Var}}(\hat{N}) = \frac{n_1^2 n_2^2 (n_2 - m)}{m^3 (n_2 - 1)} \approx \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

Derivation

Define for every element $y_i = 1$, so $\tau_y = \sum_{i=1}^{N} y_i = N$, and $x_i = 1$ if element i is marked, 0

otherwise, so $\tau_x = \sum_{i=1}^{N} x_i = n_1$ is known. The recapture sample of size n_2 gives

$$r = \frac{\sum_{i=1}^{n_2} y_i}{\sum_{i=1}^{n_2} x_i} = \frac{n_2}{m}, \qquad \hat{N} = \hat{\tau}_y = r\tau_x = \frac{n_2}{m} n_1.$$

From Ch. 5, $\widehat{\mathrm{Var}}(\hat{N}) = \tau_x^2 \widehat{\mathrm{Var}}(r) = n_1^2 \widehat{\mathrm{Var}}(r)$ with

$$\widehat{\mathrm{Var}}(r) = \frac{1}{\bar{x}^2} \cdot \frac{N - n_2}{N} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (y_i - r x_i)^2,$$

and we take the fpc ≈ 1. Now

$$\bar{x} = \frac{m}{n_2}, \qquad \sum_{i=1}^{n_2} (y_i - r x_i)^2 = \sum y_i^2 - 2r \sum x_i y_i + r^2 \sum x_i^2 = n_2 - 2\frac{n_2}{m} m + \frac{n_2^2}{m^2} m = \frac{n_2(n_2 - m)}{m},$$

using $y_i^2 = y_i = 1$ and $x_i^2 = x_i$. Therefore

$$\widehat{\mathrm{Var}}(r) = \frac{n_2^2}{m^2} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \cdot \frac{n_2(n_2 - m)}{m} = \frac{n_2^2 (n_2 - m)}{m^3 (n_2 - 1)},$$

and multiplying by n_1^2 gives the result.

1.2 Maximum likelihood

Definition 10.3

Let $f(\mathbf{x} \mid \theta)$ be the joint pdf of the sample $\mathbf{X} = (X_1, \ldots, X_n)$. Given $\mathbf{X} = \mathbf{x}$, the function $L(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta)$ of θ is the **likelihood function**. If $L(\theta_1 \mid \mathbf{x}) \geq L(\theta_2 \mid \mathbf{x})$, then θ_1 is more plausible than θ_2 in light of the data. The **maximum likelihood estimator** is $\hat{\theta}(\mathbf{x}) = \arg\max_{\theta \in \Theta} L(\theta \mid \mathbf{x})$.

Example 10.4: Normal mean

For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$,

$$\ell(\theta) = \log L(\theta \mid \mathbf{x}) = -\frac{n}{2} \log 2\pi - \frac{1}{2} \sum_{i=1}^{n} (x_i - \theta)^2, \qquad \ell'(\hat{\theta}) = \sum (x_i - \hat{\theta}) = 0 \;\Rightarrow\; \hat{\theta} = \bar{x}.$$

Theorem 10.5: Large-sample properties of the MLE

Under regularity conditions,

1. (consistency) $\lim_{n \to \infty} P(|\hat{\theta}_n - \theta| \geq \varepsilon) = 0$ for every $\varepsilon > 0$;
2. (asymptotic normality) $\sqrt{n}(\hat{\theta}_n - \theta) \to N\big(0, 1/I(\theta)\big)$, i.e. $\hat{\theta}_n \approx N\big(\theta, \frac{1}{nI(\theta)}\big)$, where $I(\theta) = \mathrm{E}\big[\big(\frac{\partial}{\partial \theta} \log f(X \mid \theta)\big)^2\big]$ is the Fisher information of one observation.

Theorem 10.6: MLE of N

Model the recapture sample as $X_i = 1$ (marked) or 0, $X_i \overset{\text{iid}}{\sim} \mathrm{Ber}(p)$ with $p = n_1/N$, $i =$

$1, \dots, n_2$. Then

$$\hat{N}_{\mathrm{MLE}} = \frac{n_1 n_2}{m}, \qquad \mathrm{Var}(\hat{N}_{\mathrm{MLE}}) \approx \frac{N^2(N-n_1)}{n_1 n_2}, \qquad \widehat{\mathrm{Var}}(\hat{N}_{\mathrm{MLE}}) = \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

Derivation

With $\sum_{i=1}^{n_2} X_i = m$,

$$L(N) = \prod_{i=1}^{n_2} \Big(\frac{n_1}{N}\Big)^{X_i} \Big(1 - \frac{n_1}{N}\Big)^{1-X_i} = \Big(\frac{n_1}{N}\Big)^{m} \Big(1 - \frac{n_1}{N}\Big)^{n_2 - m},$$

$$\ell(N) = m \log n_1 - m \log N + (n_2 - m)\log(N - n_1) - (n_2 - m)\log N, \qquad \ell'(N) = \frac{n_2 - m}{N - n_1} - \frac{n_2}{N}.$$

Setting $\ell'(N) = 0$: $N(n_2 - m) = n_2(N - n_1)$, so $\hat{N} = n_1 n_2 / m$.

Fisher information. For one observation,

$$\frac{\partial}{\partial N} \log f(x \mid N) = \frac{1 - x}{N - n_1} - \frac{1}{N} = \frac{n_1/N - x}{N - n_1} = \frac{p - x}{N - n_1},$$

$$I(N) = \mathrm{E}\Big(\frac{p - X}{N - n_1}\Big)^2 = \frac{p(1-p)}{(N-n_1)^2} = \frac{n_1}{N^2(N - n_1)}.$$

Hence $\mathrm{Var}(\hat{N}_{\mathrm{MLE}}) \approx \big[n_2 I(N)\big]^{-1} = \frac{N^2(N-n_1)}{n_1 n_2}$. Plugging in $\hat{N} = n_1 n_2 / m$:

$$\frac{(n_1 n_2/m)^2 \, (n_1 n_2/m - n_1)}{n_1 n_2} = \frac{n_1^2 n_2^2}{m^2} \cdot \frac{n_1(n_2 - m)}{m} \cdot \frac{1}{n_1 n_2} = \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

The ratio-estimator and likelihood approaches agree: $\widehat{\mathrm{Var}}(\hat{\tau}_y) \approx \widehat{\mathrm{Var}}(\hat{N}_{\mathrm{MLE}})$.

Example 10.7: Illustration

If $n_1 = 200$ fish are marked and a second catch of $n_2 = 100$ contains $m = 20$ marked fish, then $\hat{N} = 200 \cdot 100/20 = 1000$ and

$$\widehat{\mathrm{Var}}(\hat{N}) = \frac{200^2 \cdot 100 \cdot 80}{20^3} = 40{,}000, \qquad 2\sqrt{40000} = 400,$$

so N is estimated as 1000 ± 400.

2 Nonresponse

Nonresponse (missing data) biases the estimation of population parameters. Divide the population into a response stratum R and a nonresponse stratum:

	size	mean
Respondents	N_1	μ_1
Nonrespondents	N_2	μ_2
	$N = N_1 + N_2$	μ

Theorem 10.8: Nonresponse bias

If $\bar{y}_1$ is unbiased for the respondents' mean μ_1, then as an estimator of μ,

$$\mathrm{E}(\bar{y}_1) - \mu = \frac{N_2}{N}(\mu_1 - \mu_2).$$

Proof. $\mu = \frac{N_1}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \frac{N_1+N_2}{N}\mu_1 - \frac{N_2}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \mu_1 - \frac{N_2}{N}(\mu_1 - \mu_2)$. ■

The bias is small only if the nonresponse rate N_2/N is small or respondents and nonrespondents are alike ($\mu_1 \approx \mu_2$). Increasing the sample size does not reduce it.

2.1 A conservative interval for a proportion

Suppose n_1 respondents are an SRS from the N_1 respondents of the population, so a 95% interval for their proportion p_1 is

$$\hat{p}_1 \pm 2\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1 - 1} \cdot \frac{N_1 - n_1}{N_1}}.$$

Nothing is known about the nonrespondents' proportion p_2, except $0 \le p_2 \le 1$. With $w_1 = N_1/N$ and $w_2 = N_2/N$, $p = w_1p_1 + w_2p_2$. Take the worst cases $p_2 = 0$ for the lower limit and $p_2 = 1$ for the upper limit:

$$\hat{p}_L = w_1\left(\hat{p}_1 - 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\right) + w_2 \cdot 0,$$
$$\hat{p}_U = w_1\left(\hat{p}_1 + 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\right) + w_2 \cdot 1.$$

Whenever the interval for p_1 covers p_1, $[\hat{p}_L, \hat{p}_U]$ covers p whatever p_2 is, so $P(\hat{p}_L \le p \le \hat{p}_U) \ge 0.95$. The width grows by w_2: a high nonresponse rate makes the interval uninformative, which is why reducing nonresponse (call-backs, follow-ups) matters more than increasing n.

3 Selected exercises

Motivation. The Lincoln–Petersen estimator was obtained from a ratio argument and from a Bernoulli likelihood. The items below view it as the completion of a 2×2 table, derive it from the exact hypergeometric likelihood, repair its small-sample bias, and show how call-backs remove nonresponse bias.

Exercise 10.1 (Deer population)

A random sample of $n_1 = 300$ deer is captured, tagged and released. A second sample of $n_2 = 200$ taken two weeks later contains $m = 62$ tagged deer.
(a) Arrange the data as a 2×2 table (caught or not in the first and in the second sample) and estimate the missing cell, assuming that the two captures are independent.
(b) Show that the resulting estimate of N is the Lincoln–Petersen estimator, and place a bound on the error.

Solution.

(a) The table has one unobservable cell x (deer caught in neither sample):

	in 2nd sample	not in 2nd	total
in 1st sample	62	238	300
not in 1st	138	x	
total	200		N

Independence of the two captures means that the odds ratio of the table is 1, i.e. $62\,x = 238 \times 138$, so $\hat{x} = 32{,}844/62 = 529.7$.

(b)

Step 1. In general, with cells $n_{11} = m$, $n_{12} = n_1 - m$, $n_{21} = n_2 - m$ and $\hat{x} = n_{12}n_{21}/n_{11}$,

$$\hat{N} = n_{11} + n_{12} + n_{21} + \frac{n_{12}n_{21}}{n_{11}} = \frac{(n_{11}+n_{12})(n_{11}+n_{21})}{n_{11}} = \frac{n_1 n_2}{m}.$$

Step 2. Here $\hat{N} = 62 + 238 + 138 + 529.7 = 967.7 = 300 \times 200/62$.

Step 3. By the MLE theorem, $\widehat{\mathrm{Var}}(\hat{N}) = \dfrac{n_1^2 n_2 (n_2 - m)}{m^3} = \dfrac{300^2 \cdot 200 \cdot 138}{62^3} = 10{,}423$, and $2\sqrt{10{,}423} = 204$. The herd is estimated at 968 ± 204 deer. ■

Exercise 10.2 (The hypergeometric likelihood)

If the second sample is an SRS of n_2 drawn without replacement from N animals of which n_1 are marked, then

$$L(N) = P(m \mid N) = \frac{\binom{n_1}{m}\binom{N-n_1}{n_2-m}}{\binom{N}{n_2}}, \qquad N \geq n_1 + n_2 - m.$$

Assuming $m \geq 1$, show that the MLE of N over the integers is $\lfloor n_1 n_2 / m \rfloor$.

Solution. [Supplement]

Step 1. (Likelihood ratio.) For $N - 1 \geq n_1 + n_2 - m$, using $\binom{a}{c}/\binom{a-1}{c} = \frac{a}{a-c}$,

$$\frac{L(N)}{L(N-1)} = \frac{\binom{N-n_1}{n_2-m}}{\binom{N-1-n_1}{n_2-m}} \cdot \frac{\binom{N-1}{n_2}}{\binom{N}{n_2}} = \frac{N-n_1}{N-n_1-n_2+m} \cdot \frac{N-n_2}{N}.$$

Step 2. (Sign.) Both denominators are positive, so $L(N) \geq L(N-1)$ iff $(N-n_1)(N-n_2) \geq N(N-n_1-n_2+m)$. Expanding, the terms N^2 and $-N(n_1+n_2)$ cancel and the condition becomes $n_1 n_2 \geq Nm$, i.e. $N \leq n_1 n_2 / m$.

Step 3. (Maximum.) Hence L is nondecreasing for $N \leq n_1 n_2/m$ and strictly decreasing afterwards, so it is maximized at $N = \lfloor n_1 n_2 / m \rfloor$ (if $n_1 n_2/m$ is an integer, $n_1 n_2/m - 1$ attains the same value). This value is at least $n_1 + n_2 - m$, since $(n_1 - m)(n_2 - m) \geq 0$ gives $n_1 n_2/m \geq n_1 + n_2 - m$. The exact likelihood therefore gives the Lincoln–Petersen estimate, as the Bernoulli approximation of the text did. ■

Exercise 10.3 (Chapman's unbiased modification)

In the setting of Exercise 10.2, let $\tilde{N} = \dfrac{(n_1+1)(n_2+1)}{m+1} - 1$.

(a) Show that $\mathrm{E}(\tilde{N}) = N$ whenever $n_1 + n_2 \geq N$.

(b) Explain why $\hat{N} = n_1 n_2/m$ has no finite expectation when $n_2 \leq N - n_1$.

Solution. [Supplement]

(a)

Step 1. Since $\frac{1}{m+1}\binom{n_1}{m} = \frac{1}{n_1+1}\binom{n_1+1}{m+1}$,

$$\mathrm{E}\Big(\frac{1}{m+1}\Big) = \frac{1}{(n_1+1)\binom{N}{n_2}} \sum_{m} \binom{n_1+1}{m+1}\binom{N-n_1}{n_2-m} = \frac{1}{(n_1+1)\binom{N}{n_2}} \sum_{k\geq 1} \binom{n_1+1}{k}\binom{N-n_1}{n_2+1-k}.$$

Step 2. By Vandermonde's identity the sum over all $k \geq 0$ is $\binom{N+1}{n_2+1}$, and the $k=0$ term is $\binom{N-n_1}{n_2+1}$, which is 0 when $n_2+1 > N-n_1$, i.e. when $n_1+n_2 \geq N$. Then

$$\mathrm{E}\Big(\frac{1}{m+1}\Big) = \frac{\binom{N+1}{n_2+1}}{(n_1+1)\binom{N}{n_2}} = \frac{N+1}{(n_1+1)(n_2+1)}.$$

Step 3. Hence $\mathrm{E}(\tilde{N}) = (n_1+1)(n_2+1)\mathrm{E}(\frac{1}{m+1}) - 1 = N$.

When $n_1+n_2 < N$ the bias is $-(N+1)\binom{N-n_1}{n_2+1}/\binom{N+1}{n_2+1}$, which is negligible unless $P(m=0)$ is appreciable.

(b) If $n_2 \leq N-n_1$, the second sample can consist of unmarked animals only, so $P(m=0) = \binom{N-n_1}{n_2}/\binom{N}{n_2} > 0$ and $\hat{N} = \infty$ on that event. Adding 1 to the counts, as in $\tilde{N}$, removes this defect. ■

Exercise 10.4 (Call-backs for nonrespondents)

An SRS of size n is drawn; n_1 units respond and $n_2 = n - n_1$ do not. If $n_2 \geq 1$, an SRS of r of the n_2 nonrespondents ($1 \leq r \leq n_2$, where r may depend on n_2) is followed up by an intensive call-back and all of them respond; if $n_2 = 0$ there is no call-back. Let $\bar{y}_1$ be the mean of the first-round respondents, $\bar{y}_{2r}$ the mean of the call-back subsample, and

$$\bar{y}_{HH} = \frac{n_1\bar{y}_1 + n_2\bar{y}_{2r}}{n}.$$

Show that $\bar{y}_{HH}$ is unbiased for μ, whatever the difference between respondents and nonrespondents.

Solution. [Supplement]

Step 1. Fix the first-phase sample s; it determines n_1, n_2, the respondents and the nonrespondents. The call-back subsample is an SRS of the n_2 sampled nonrespondents, so by Ch. 2 $\mathrm{E}(\bar{y}_{2r} \mid s) = \bar{y}_2$, the mean of all n_2 nonrespondents in s. (If $n_2 = 0$ the second term of $\bar{y}_{HH}$ is absent and $\bar{y}_{HH} = \bar{y}_1 = \bar{y}$.)

Step 2. Hence $\mathrm{E}(\bar{y}_{HH} \mid s) = \frac{n_1\bar{y}_1 + n_2\bar{y}_2}{n} = \bar{y}$, the mean of the whole first-phase sample (which could not be computed).

Step 3. By iterated expectation and Ch. 2, $\mathrm{E}(\bar{y}_{HH}) = \mathrm{E}(\bar{y}) = \mu$.

The device works because the population is split into the two strata of the nonresponse theorem and the nonresponse stratum is represented by a random subsample. Unlike $\bar{y}_1$, whose bias $\frac{N_2}{N}(\mu_1 - \mu_2)$ does not shrink with n, $\bar{y}_{HH}$ is unbiased at the price of extra variance from subsampling. ■